

1. Three particles of masses m , $3m$ and km are placed at the points whose coordinates are $(2, 1)$, $(4, 3)$ and $(1, 5)$ respectively. The centre of mass of the three particles lies on the straight line with equation $y = 2x$.

Find the value of k .

(6)

2. A van has mass 750 kg. The van is moving up a straight road at constant speed $U \text{ m s}^{-1}$

The road is inclined to the horizontal at an angle ϑ , where $\sin \vartheta = \frac{1}{15}$

The resistance to the motion of the van from non-gravitational forces is modelled as a single force of magnitude $30U$ newtons. The engine of the van is working at a constant rate of 16 kW.

Find the value of U .

3. A particle, Q , of mass 2 kg is moving under the action of a single force $\mathbf{F}$ newtons. At time t seconds ($t \geq 0$), the position vector of Q , relative to a fixed point O , is $\mathbf{r}$ metres and the velocity of Q is $\mathbf{v}\text{ ms}^{-1}$ where

$$\mathbf{v} = (2t + 10)\mathbf{i} + 9t^{\frac{1}{2}}\mathbf{j}$$

Given that $\mathbf{r} = -20\mathbf{j}$ when $t = 0$, find

- (a) $\mathbf{F}$ when $t = 4$ (3)
- (b) $\mathbf{r}$ when $t = 4$ (3)
- (c) the values of t for which Q is moving in the direction $\mathbf{i} + \mathbf{j}$. (3)

4.

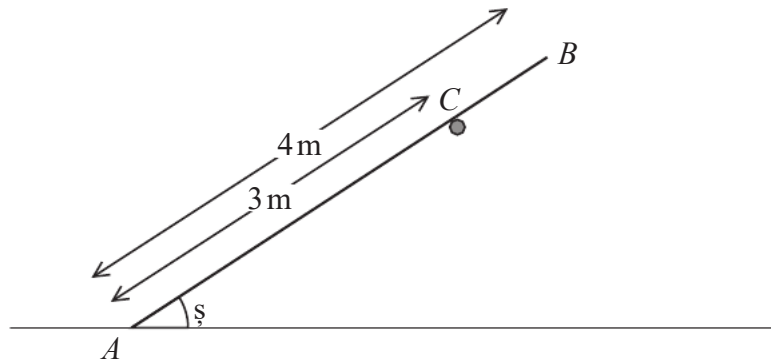


Figure 1

A uniform rod, AB , has length 4 m and weight 160 N. The rod rests against a fixed small smooth horizontal peg. The peg is perpendicular to the vertical plane containing AB . The rod rests on the peg at C , where $AC = 3$ m. The end A of the rod rests on rough horizontal ground and the rod is at an angle φ to the ground, as shown in Figure 1.

Given that $\cos \varphi = \frac{3}{4}$ and the rod is in limiting equilibrium,

(a) show that the magnitude of the force acting on the rod at C is 80 N,

(3)

(b) find the coefficient of friction between the rod and the ground.

(6)

5.

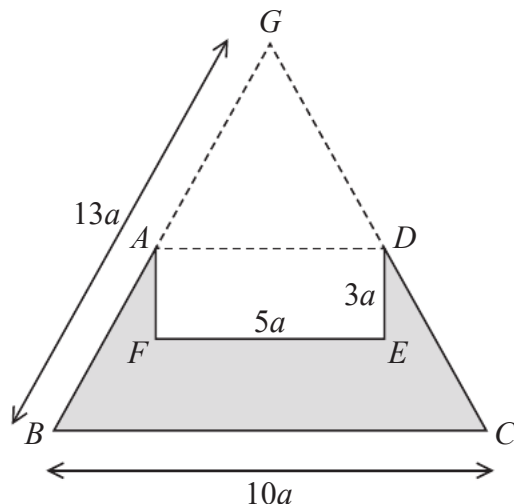


Figure 2

The uniform lamina GBC is in the shape of an isosceles triangle with $GB = GC = 13a$ and $BC = 10a$. The midpoint of GB is A and the midpoint of GC is D . The rectangle $AFED$ is such that $FE = 5a$ and $DE = 3a$.

The template, shown shaded in Figure 2, is formed by removing the triangle GAD and the rectangle $ADEF$ from the lamina GBC .

- (a) Show that the centre of mass of the template is $\frac{7}{4}a$ from BC .

(5)

The template is freely suspended from A and hangs in equilibrium with AB at an angle of $\S^\circ$ to the downward vertical.

- (b) Find the value of $\S$, giving your answer to the nearest whole number.

(5)

6.

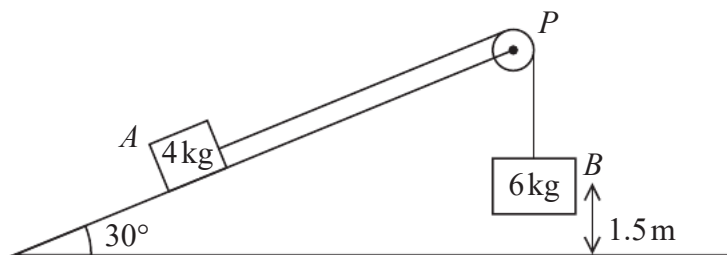


Figure 3

Two blocks, A and B , of masses 4 kg and 6 kg respectively, are connected by a light inextensible string. Block A is held at rest on a fixed rough ramp that is inclined at an angle of 30° to the horizontal. The string passes over a small smooth pulley, P , which is fixed at the top of the ramp. Block B hangs vertically below P , 1.5 m above the ground, as shown in Figure 3. Block A is more than 1.5 m from P . The blocks are released from rest with the string taut so that A moves up the ramp and the section of the string from A to P is parallel to a line of greatest slope of the ramp. The coefficient of friction between A and the ramp is $\frac{1}{\sqrt{3}}$.

The blocks are modelled as particles and air resistance is ignored.

- (a) Find the work done against friction as A moves 1.5 m up the ramp. (4)
- (b) Find the potential energy lost by the system as A moves 1.5 m up the ramp. (3)
- (c) Use the work-energy principle to find the speed of B at the instant immediately before it hits the ground. (3)

7. Particles A , B and C , of masses $8m$, $2m$ and $3m$ respectively, lie at rest in a straight line on a smooth horizontal plane with B between A and C .

Particles B and C are projected towards each other with speeds ku and $4u$ respectively so that they collide directly. Immediately after the collision, the speed of B is $2u$ and the speed of C is u , and, as a result of the collision, the direction of motion of each of B and C is reversed.

The total kinetic energy lost in the collision between B and C is $3PX^2$, where 3 is a constant.

- (a) Find the value of 3 .

(5)

After the collision between B and C , particle B , moving with speed $2u$, collides directly with particle A . The coefficient of restitution between A and B is e .

Given that there is a second collision between B and C ,

- (b) find the range of possible values of e .

(6)

8. [In this question, the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.]

Two fixed points, X and Y , lie on horizontal ground with $XY = 60\text{ m}$. At time $t = 0$ a particle, P , is projected from X with velocity $(12\mathbf{i} + b\mathbf{j})\text{ m s}^{-1}$. At the same instant, another particle, Q , is projected from Y with velocity $(-c\mathbf{i} + d\mathbf{j})\text{ m s}^{-1}$.

Given that the particles collide at a point Z and that b , c and d are positive constants,

- (a) show that $b = d$.

(1)

The point Z is 10 m above the horizontal ground and the collision occurs at time $t = 3$ seconds.

- (b) Find the value of d .

(2)

- (c) Find the length of time for which Q is at least 10 m above the ground.

(3)

- (d) Find the direction of motion of Q at the instant before it collides with P .

(5)

At time $t = T$ seconds, the direction of motion of P is perpendicular to its direction of motion at time $t = 0$

- (e) Find the value of T .

(4)